

01-11-2025

Agenda:

- Naive Bayes Algo (2 methods)
- k-means clustering

Naive Bayes algorithm

Bayes theorem :
$$p(y|x) = \frac{p(x|y) \cdot p(y)}{p(x)}$$

↓
probability

what if?

- data is categorical?
- we want probabilistic reasoning
- we want something that learns from frequency count.
- we want fast & interpretable solution.
- NB → classification → probabilistic model

Data set: (play Tennis)

Outlook

Temperature

play

prob of Rainy & Hot ?

Sunny	Hot	No
Sunny	Mild	No
cloudy	Hot	Yes
Rainy	Mild	Yes
Rainy	cool	Yes
cloudy	cool	Yes

$$p(\text{Yes}) = \frac{4}{6} \rightarrow 0.66$$

$$p(\text{No}) = \frac{2}{6} \rightarrow 0.33$$

$p(\text{Yes}) \rightarrow 0.66$	outlook	O - calc	Temp	T - calc
	cloudy	$2/4 \rightarrow 0.5$	Hot	$1/4 \rightarrow 0.25$
	Rainy	$2/4 \rightarrow 0.5$	Mild	$1/4 \rightarrow 0.25$
	Sunny	$0/4 \rightarrow 0$	cool	$2/4 \rightarrow 0.5$
$p(\text{No}) \rightarrow 0.33$	cloudy	$0/2 \rightarrow 0$	Hot	$1/2 \rightarrow 0.5$
	Rainy	$0/2 \rightarrow 0$	Mild	$1/2 \rightarrow 0.5$
	Sunny	$2/2 \rightarrow 1$	cool	$0/2 \rightarrow 0$

→ prob of Rainy & Hot

$$\begin{aligned}
 p(\text{Yes} | \text{Rain, Hot}) &\propto p(\text{Yes}) \times p(\text{Rain} | \text{Yes}) \times p(\text{HOT} | \text{Yes}) \\
 &= 0.66 \times 0.5 \times 0.25 \\
 &= 0.0825
 \end{aligned}$$

$$\begin{aligned}
 p(\text{No} \mid \text{Rain}, \text{Hot}) &\propto p(\text{No}) \times p(\text{Rain} \mid \text{No}) \times p(\text{Hot} \mid \text{No}) \\
 &= 0.33 \times 0 \times 0.5 \\
 &= 0
 \end{aligned}$$

$$p(\text{Yes} \mid \text{Rain}, \text{Hot}) > p(\text{No} \mid \text{Rain}, \text{Hot}) \Rightarrow \text{Yes}$$

→ prob of Sunny & cool

$$\begin{aligned}
 p(\text{Yes} \mid \text{Sunny}, \text{cool}) &\propto p(\text{Yes}) \times p(\text{Sunny} \mid \text{Yes}) \times p(\text{cool} \mid \text{Yes}) \\
 &= 0.66 \times 0 \times 0.5 \\
 &= 0
 \end{aligned}$$

$$\begin{aligned}
 p(\text{No} \mid \text{Sunny}, \text{cool}) &\propto p(\text{No}) \times p(\text{Sunny} \mid \text{No}) \times p(\text{cool} \mid \text{No}) \\
 &= 0.33 \times 1 \times 0 \\
 &= 0
 \end{aligned}$$

$$p(\text{Yes} \mid \text{Sunny}, \text{cool}) \quad \& \quad p(\text{No} \mid \text{Sunny}, \text{cool})$$

Both are 0

↓
To solve this we need smoothing

we add 1 to every count & divide by
samples + # categories

outlook o - calc			Temp T - calc		
cloudy	2/4 →	2+1/4+3 → 0.429	Hot	1/4 →	1+1/4+3 → 0.286
Rainy	2/4 →	2+1/4+3 → 0.429	Mild	1/4 →	1+1/4+3 → 0.286
Sunny	0/4 →	0+1/4+3 → 0.143	Cool	2/4 →	2+1/4+3 → 0.429
cloudy	0/2 →	0+1/2+3 → 0.2	Hot	1/2 →	1+1/2+3 → 0.4
Rainy	0/2 →	0+1/2+3 → 0.2	Mild	1/2 →	1+1/2+3 → 0.4
Sunny	2/2 →	2+1/2+3 → 0.6	Cool	0/2 →	0+1/2+3 → 0.2

$$p(Yes) \rightarrow 4/6 \rightarrow 0.66$$

$$p(No) \rightarrow 2/6 \rightarrow 0.33$$

→ prob of sunny & cool

$$\begin{aligned}
 p(Yes | \text{sunny, cool}) &\propto p(Yes) \times p(\text{sunny} | Yes) \times p(\text{cool} | Yes) \\
 &= 0.66 \times 0.143 \times 0.429 = \underline{\underline{0.040}}
 \end{aligned}$$

$$\begin{aligned}
 p(No | \text{sunny, cool}) &\propto p(No) \times p(\text{sunny} | No) \times p(\text{cool} | No) \\
 &= 0.33 \times 0.6 \times 0.2 \\
 &= \underline{\underline{0.0396}}
 \end{aligned}$$

Normalise to get the final probabilities:

$$\text{sum} = 0.04 + 0.0396 = 0.0796$$

$$p(\text{Yes} | \text{sunny}, 1001) = \frac{0.040}{0.0796} \approx 0.5025 \rightarrow \text{sum} \approx 1$$

$$p(\text{No} | \text{sunny}, 1001) = \frac{0.0396}{0.0796} \approx 0.4974 \rightarrow$$

$$p(\text{Yes} | \text{sunny}, 1001) > p(\text{No} | \text{sunny}, 1001)$$

$$p(\text{Yes}) = p(\text{No})$$

→ features → categorical → Categorical Naive Bayes
 → target → categorical →

Next topic:

Dataset → numeric features → categorical NB?
 → categorical target column

→ Gaussian NB
 ↓
 (normal)

we compute:

$$p(y|x) \propto p(y) \prod \mathcal{N}(x | \mu_y, \sigma_y^2)$$

$$\mathcal{N}(x, \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \cdot \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

$$\begin{array}{c} \exp(2) \\ \downarrow \\ e^2 \\ \downarrow \\ 2.718^2 \end{array}$$

Data set:

id	x_1	x_2	y
A	1.0	2.0	Yes
B	2.0	1.0	Yes
C	1.5	1.8	Yes
D	3.0	3.0	No
E	3.5	2.8	No
F	2.9	3.2	No

Step 1: Calculate MLE:

Maximum likelihood estimate: (MLE) → uses n instead of $n-1$ which is used in unbiased sample variance

mean, variance, std

$$\underline{\mu(x_1 = \text{Yes})} = 1 + 2 + 1.5 / 3 \rightarrow \mu_{\text{Yes}, 1} = 1.5$$

$$\underline{\mu(x_2 = \text{Yes})} = 2 + 1 + 1.8 / 3 \rightarrow \mu_{\text{Yes}, 2} = 1.6$$

$$\underline{\mu(x_1 = \text{No})} = 3 + 3.5 + 2.9 / 3 \rightarrow \mu_{\text{No}, 1} = 3.133$$

$$\underline{\mu(x_2 = \text{No})} = 3 + 2.8 + 3.2 / 3 \rightarrow \mu_{\text{No}, 2} = 3.0$$

Variance x_1 :

$$\begin{aligned} (\text{Yes}) &: (1 - 1.5)^2, (2 - 1.5)^2, (1.5 - 1.5)^2 \\ &: (-0.5)^2, (0.5)^2, 0^2 \end{aligned}$$

$$\text{sum} : 0.25 + 0.25 + 0 \rightarrow 0.5 / 3 \rightarrow 0.166$$

$$\sigma^2 = \sum (x - \mu)^2 / N$$

variance x_2 :

$$(Yes) : (0.4)^2, (-0.6)^2, (0.2)^2 \rightarrow 0.56$$
$$: 0.56/3 \rightarrow 0.186$$

variance x_1 : sum $\rightarrow 0.2066 \rightarrow 0.2066/3 \rightarrow 0.0688$
(No)

variance x_2 : sum $\rightarrow 0.08 \rightarrow 0.08/3 \rightarrow 0.0266$
(No)

$$x = \begin{matrix} x_1 & x_2 \\ \downarrow & \downarrow \\ (2.0, 2.0) \end{matrix}$$

Step 2: calculate gaussian prob per feature:

for class = Yes

$\rightarrow x_1$:

$$\rightarrow \sigma = \sqrt{0.166} \approx 0.408$$

$$\rightarrow \text{coeff} = 1/(\sqrt{2\pi}\sigma^2) \approx 0.9772$$

$$\rightarrow \text{exponent exp: } \left(-\frac{(2-1.5)^2}{2 \times 0.166} \right) \approx \exp(-0.75) \approx 0.472$$

$$\rightarrow p(x_1 | \underline{Yes}) \approx \frac{1}{\sqrt{2\pi}\sigma^2} \cdot \exp\left(-\frac{(x-4)^2}{2\sigma^2}\right)$$

$$\approx 0.9772 \times 0.472$$

$$\approx 0.4616$$

$$p(x_1 | y_{e1}) \checkmark \quad p(x_2 | y_{e1}) \checkmark$$

$$p(x_1 | N_0) \quad p(x_2 | N_0)$$

x_2 :

$$\rightarrow \sigma = \sqrt{0.1866} \approx 0.4320$$

$$\rightarrow \text{corr} \approx 0.923$$

$$\rightarrow \text{exp on exp} : \exp\left(-\frac{(2-1.6)^2}{2 \times 0.1866}\right) = \exp(-0.428) \approx 0.6510$$

$$p(x_2 | y_{e1}) \approx 0.4320 \times 0.6510 \\ \approx 0.6009$$

for class = No :

$$p(x_1 | N_0) \approx 1.36 \times 10^{-9}$$

$$p(x_2 | N_0) \approx 1.68 \times 10^{-8}$$

$$p(x_1 | y_{e1}) \approx 0.4616 \quad p(x_2 | y_{e1}) \approx 0.6009$$

$$\text{Joint prob!} \quad p(x | y_{e1}) = 0.4616 \times 0.6009 \\ \approx 0.2774$$

$$p(x_1 | N_0) \approx 1.36 \times 10^{-9} \quad p(x_2 | N_0) \approx 1.68 \times 10^{-8}$$

$$\text{Joint prob!} \quad p(x | N_0) \approx 2.29 \times 10^{-12}$$

$$p(x | \text{Yes}) \approx 0.2774 \quad p(x | \text{No}) \approx 2.29 \times 10^{-12}$$

↓

unnormalized prob

$$\begin{aligned} p(\text{Yes} | x) &: p(\text{Yes}) \cdot p(x | \text{Yes}) & p(\text{No} | x) &: p(\text{No}) \cdot p(x | \text{No}) \\ &: 0.5 \times 0.2774 & &: 0.5 \times 2.29 \times 10^{-12} \\ &\approx 0.1387 & &: 1.145 \times 10^{-12} \end{aligned}$$

Step 3: normalize to get class probabilities.

$$\text{Sum } S : 0.1387 + 1.145 \times 10^{-12} \approx 0.1387$$

so normalized:

$$p(\text{Yes} | x) \approx \frac{0.1387}{0.1387} \approx 0.9999 \dots \approx 1$$

$$p(\text{No} | x) \approx \frac{1.145 \times 10^{-12}}{0.1387} \approx 8.26 \times 10^{-12}$$

Decision: predict $\rightarrow$ Yes (very strongly)

↓

$$\log p(y|x) = \log p(y) + \prod \log \mathcal{N}(x | \mu, \sigma^2)$$

↑

python implement

when to use Gaussian NB:

- features are continuous
 - you want a very fast, interpretable model,
 - we are estimating that features are normally distributed,
-

email $\rightarrow$ spam $\rightarrow$
 $p(s) = 0.2$
Not spam
 $p(n) = 0.8$

J : please click on the link below

$$p(\text{open} | J) = x_1$$

$$p(\text{click} | J) = x_2$$

$$n) : p(\quad) = \frac{p_1}{p_1 + p_2}$$

→ Tomorrow is holiday
 $\downarrow$
encode

$$p(\text{Yes} | x) = p(\text{spam}) \times p(\text{Tom} | \text{Yes}) \times p(\text{$$

$$= 0.6$$

$$p(\text{No} | x) = 0.4$$

New Algo.

k - mean clustering !

→ unsupervised learning → No target column

Data → k-mean clustering → group data based on similarity (clustering)

↑
how many
group you
want?

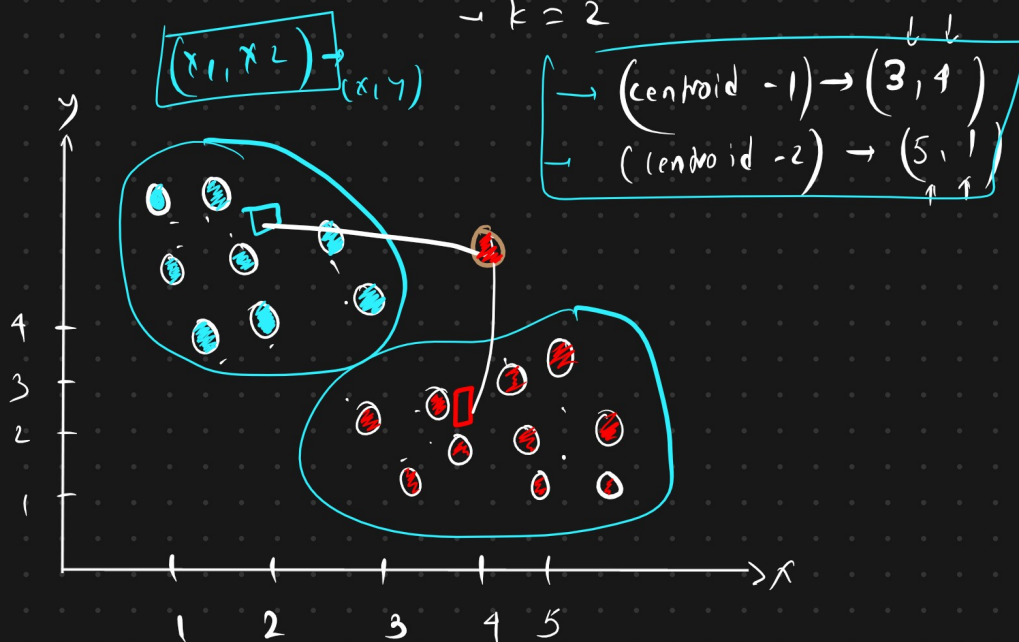
↑
K

→ K → no of centroids
→ K = 2

Initiation :

Stop train

B → 8
R → 10
Iter - 3 :
B → 8
R → 10



Data set:

$$A = (1, 2)$$

$$B = (1.5, 1.8)$$

$$C = (5, 8)$$

$$D = (8, 8)$$

$$E = (1, 0.6)$$

$$F = (9, 11)$$

k=2

$$c_1 = A = (1, 2)$$

$$c_2 = C = (5, 8)$$

Euclidean distance:

$$(x_1, x_2), (A_1, A_2)$$

$$\sqrt{(x_1 - A_1)^2 + (x_2 - A_2)^2}$$

Iteration 1: (Step 1)

1. point A (1, 2)

$$\text{to } c_1: \sqrt{(1-1)^2 + (2-2)^2} = 0$$

$$\text{to } c_2: \sqrt{(1-5)^2 + (2-8)^2} = 7.211$$

→ assign to cluster c1

2. point B (1.5, 1.8)

$$\text{to } c_1: \sqrt{(1.5-1)^2 + (1.8-2)^2} \approx 0.5385$$

$$\text{to } c_2: \sqrt{(1.5-5)^2 + (1.8-8)^2} \approx 7.119$$

→ cluster 1

3. point C → cluster 2

4. point D → cluster 2

5. point E → cluster 1

6. point F → cluster 2

cluster 1: { A, B, E }

cluster 2: { C, D, F }

Step 2: update centroids

New c1: mean of A(1, 2), B(1.5, 1.8), E(1, 0.6)

$$c1-x: (1 + 1.5 + 1) / 3 = 1.1667$$

$$c1-y: (2 + 1.8 + 0.6) / 3 = 1.4667$$

New c2:

mean of C(5, 8), D(8, 8), F(9, 11)

$$c2-x: (5 + 8 + 9) / 3 \approx 7.33$$

$$c2-y: (8 + 8 + 11) / 3 \approx 9$$

$$c1: (1.1667, 1.4667)$$

$$c2: (7.33, 9)$$

How to compute loss $\rightarrow$ WCSS

$\rightarrow$ we calculate WCSS after every iteration of centroid updates.

$\rightarrow$ WCSS $\rightarrow$ within cluster summation square

$\rightarrow$ Sum over cluster & point of squared distance to their centroid.

Iteration: 2

$\rightarrow A(1,2) \approx 0.5587$

$\rightarrow c_1$

$\rightarrow B(1.5, 1.8) \approx 0.4714$

$\rightarrow c_1$

$\rightarrow C(5,8) \approx 2.539$

$\rightarrow c_2$

$\rightarrow D(8,8) \approx 1.2019$

$\rightarrow c_2$

$\rightarrow E(9,11) \approx 2.602$

$\rightarrow c_2$

new- $c_1 (1.1667, 1.4667)$

new- $c_2 (7.3333, 9)$

WCSS :

$c_1 \rightarrow A, B, E$

$$(0.5587)^2 + (0.4714)^2 + (0.1837)^2$$

$$= 1.3163$$

$c_2 \rightarrow C, D, E$

$$\rightarrow (2.539)^2 + (1.2019)^2 + (2.602)^2$$

$$\rightarrow 14.666$$

$$WCSS \rightarrow WCSS(c_1) + WCSS(c_2)$$

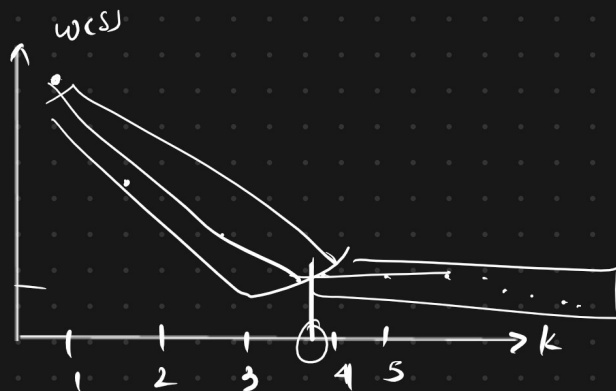
$$\rightarrow \underline{15.982}$$

$$\underline{18.683}$$

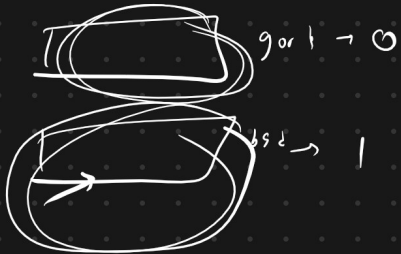
$\rightarrow$ To get optimal value k , plot elbow graph

x-axis $\rightarrow k$

y-axis $\rightarrow$ WCSS



increase in k, decrease WCSS

dataset $\rightarrow$ 1000 rows $\rightarrow$ kmean (2) $\rightarrow$  $\rightarrow$ 0

bad good $\rightarrow$ k=2

